

How do Different Diseases Spread, and how does Network Science Help us Protect Populations?

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ABSTRACT

How an epidemic spreads, and how best to contain it, depends jointly on the disease and on the transport network it travels, two factors that are usually studied apart. Here we bring them together on a realistic multimodal substrate, as a controlled and fully reproducible comparison, delivered as a pipeline with an interactive companion app and scalable to any region. We build a shared-node multilayer metapopulation of Europe, coupling air as diffusive relocation, land as recurrent commuting, and water as ferries, and run five disease types parameterised from the literature, from an immunizing measles-like SIR to a slow endemic gonorrhea-like SIS. Holding all but the studied factor fixed, we find that targeting bridges (betweenness) consistently beats targeting hubs (degree) across all five types, and by a wider margin as the network grows more realistic, while grounding every flight only halves the outbreak, because land and ferry travel keep carrying it.

1. INTRODUCTION

The spatial spread of infectious diseases depends both on pathogen biology and on the structure of human mobility. Characteristics such as transmissibility, infectious period, latency, mortality, reinfection, or waning immunity determine how an outbreak evolves locally, while transport infrastructures determine how infections move between populations. Air, sea, and land transport systems have increased the reach, speed, and volume of human movement, making mobility networks central to the geographical dissemination of infectious diseases [42]. In highly connected regions such as Europe, local outbreaks can therefore become regional or continental, with air travel supporting rapid long-distance seeding [13] and multiscale mobility processes, including commuting and long-range travel, shaping spatial and temporal spreading patterns [4]. Because these corridors carry very different volumes and operate on very different timescales, the same pathogen can produce markedly different geographical footprints depending on which parts of the transport system it reaches first. Understanding this interaction between disease biology and mobility structure is a prerequisite for anticipating where an outbreak will travel and for deciding where limited protective effort is best spent [10].

Mathematical epidemic models provide a controlled way to compare these mechanisms. Classical compartmental mod-

els divide the population into epidemiological states, beginning with the SIR framework of Kermack and McKendrick [24], in which susceptible individuals become infected and are then removed or recovered. Extensions represent different disease archetypes: SEIR adds a latent exposed state, SIS represents infections without lasting immunity, SEIRS allows immunity to wane, and SEIQRD includes quarantine and disease-induced death [37]. Such models are widely used to analyse transmission mechanisms, epidemic thresholds, and control strategies [22]. However, homogeneous compartmental models describe only the dynamics within a single population. To model spatial spread between cities, this work uses a metapopulation approach, where each city is a local population with internal epidemic dynamics and mobility flows connect cities, allowing infections to be transported across the network [16].

Figure 1 summarizes this modelling logic by combining within-city epidemic dynamics with mobility flows between cities.

Europe is represented here as a multilayer transport network composed of air, water, and land mobility layers. Multilayer network theory is appropriate for systems in which the same set of nodes can be connected by different types of links, each corresponding to a distinct interaction channel or transport mode [26; 9]. In this work, cities are nodes and transport routes are organised by layer: air links represent fast long-distance mobility, water links represent maritime and ferry-based connectivity, and land links represent regional travel and commuting. This distinction is epidemiologically relevant because mobility can operate through different mechanisms. Air and water travel are approximated as more diffusive, meaning that individuals may relocate and mix in a destination city, whereas land commuting is recurrent, meaning that individuals travel, mix temporarily, and return to their home city. Recurrent mobility can synchronize nearby cities, while long-distance diffusive mobility can seed infection in distant populations [7; 4].

Figure 2 illustrates the conceptual difference between these two mobility mechanisms.

Beyond mobility volume, the topology of the network influences epidemic propagation. Cities and routes are not structurally equivalent: some cities act as hubs with many connections, others as bridges between regions, and others belong to dense structural cores that may sustain spreading processes. Epidemic processes in complex networks are strongly affected by heterogeneity, connectivity, and node position [35]. Moreover, geographical distance is not always the best proxy for epidemiological proximity, since strongly

Metapopulation epidemic framework

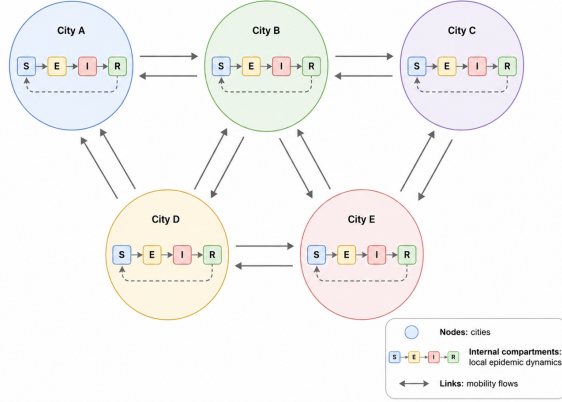


Figure 1: Metapopulation epidemic framework. Each circle represents a city, modelled as a local subpopulation with its own compartmental epidemic dynamics. The internal compartments illustrate a generic SEIR-type progression: S denotes susceptible individuals, E exposed individuals who are infected but not yet infectious, I infectious individuals, and R recovered or removed individuals. The dashed return arrow indicates that some models, such as SIS or SEIRS, allow individuals to become susceptible again. Arrows between cities represent mobility flows that can transport susceptible, exposed, or infected individuals and seed outbreaks elsewhere. Inspired by metapopulation epidemic models such as Colizza and Vespignani [16].

connected cities can be effectively close even when they are geographically distant [10].

Network science metrics can therefore support intervention design. Degree identifies the highest-traffic cities, betweenness centrality identifies the cities that bridge regions, and dense-neighbourhood measures such as triangle participation identify cities embedded in tightly connected cores that can sustain spreading [25; 31]. More recent optimal-percolation methods, notably Collective Influence and non-backtracking centrality, go further by targeting the low-degree, high-impact bridge cities that raw degree overlooks [32; 38]. These metrics can guide node interventions, interpreted as vaccination, protection, screening, or intensified surveillance in selected cities, and edge interventions, interpreted as closure or restriction of selected routes. Targeted immunization and edge-removal strategies can outperform random interventions in heterogeneous networks, although their effectiveness depends on the network structure and the timing and intensity of control [36; 3; 45]. The objective of this work is comparative rather than predictive: the aim is not to reproduce a specific European epidemic, but to compare how compartmental models, mobility layers, and intervention strategies shape simulated spreading. The central question is how the multilayer structure of European transport enables or limits epidemic processes, and how network metrics can guide containment. Combining compartmental models, a metapopulation framework, multilayer networks, and topology-based interventions, the study systematically relates human mobility, network structure, and the spatial dynamics of infectious disease spread.

Mobility mechanisms

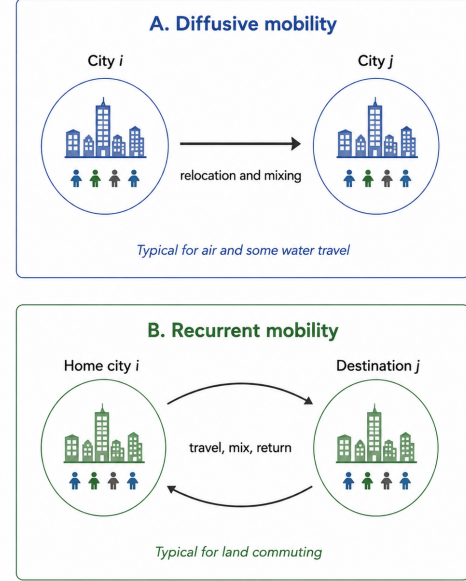


Figure 2: Diffusive and recurrent mobility mechanisms. In diffusive mobility, individuals move from city i to city j , relocate, and mix with the destination population; this approximates air travel and some water-based connections. In recurrent mobility, individuals travel from a home city i to a destination j , mix temporarily, and return home; this represents land commuting and regional travel. Diffusive mobility can seed distant outbreaks, whereas recurrent mobility mainly couples and synchronizes nearby cities [7; 4].

2. METHODOLOGY

Our question is how different disease types spread across the transport network that connects Europe, and which network-science protection best contains them. Answering it fairly requires three components built in sequence: a transport network from open data (Section 2.1), a metapopulation epidemic engine that runs on it (Section 2.2), and the disease types and protection strategies evaluated on top (Sections 2.3 and 2.4). One principle constrains all of them: in any comparison, only the factor under study varies while everything else is held fixed. The study is therefore comparative rather than predictive: we do not estimate how many people a real epidemic would infect, but quantify differences between scenarios under identical conditions. The whole pipeline, from data retrieval through network generation to evaluation, is a single reproducible workflow (Figure 3).

2.1 Building the map

The substrate is a weighted graph $G = (V, E)$ in which nodes $v \in V$ are cities and edges $(i, j) \in E$ are travel routes carrying weight w_{ij} proportional to traffic. We construct G with NetworkX [20] from four open sources: air routes from OpenFlights [34], city populations from GeoNames, road and rail topology from OpenStreetMap and GRIP [30], and ferry routes from OpenStreetMap. Region is a parameter of construction, which lets us hold the construction method fixed

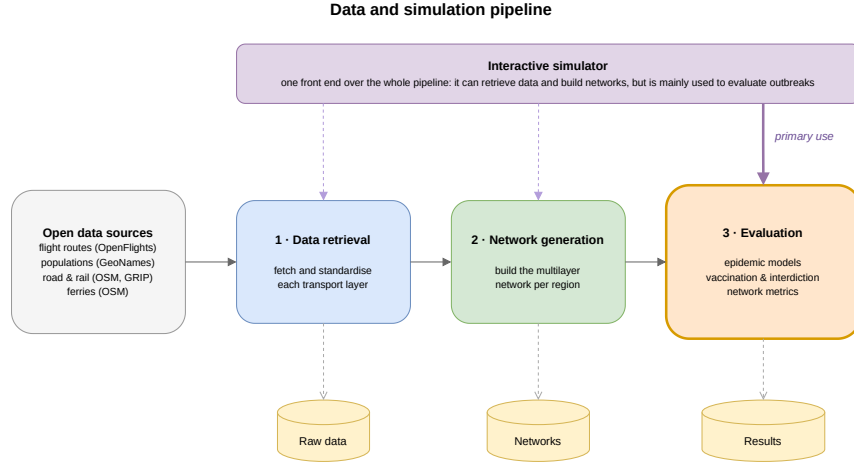


Figure 3: Data and simulation pipeline. Open data sources are retrieved, assembled into per-region multilayer networks, and evaluated by one engine. A single interactive simulator drives the whole pipeline and can retrieve data and build networks, but is mainly used to evaluate outbreaks.

while varying only geography.

Because a single air layer misrepresents how an epidemic actually propagates, the same node set carries up to three layers, air, land, and water, combined into one shared-node multilayer network (Figure 4). Edge weights are layer-specific passenger-volume proxies, but they come from different recipes according to what data exists. Air and water weights are real route frequencies: OpenFlights for air, and an OpenStreetMap ferry dump for water, filtered to passenger-carrying services (OSM tag `passenger` not equal to `no`) over crossings at least 5 km long, which drops freight-only and short canal hops while keeping genuine straits and sea ferries. Land, which has no comparably uniform cross-region flow data, is instead modelled with a radiation kernel $K_{ij} = \text{flux}_{ij}/\text{pop}_i$ [39] computed from populations. Fairness here requires not that all layers share one recipe, but that each layer is built by a single method applied identically to every region, so that differences between regions reflect topology rather than the data source; European commuting data (Eurostat) is reserved for validating the land kernel, not for parameterising any region. Node populations N_i come from GeoNames where available, with a degree-based proxy $N_i = P_0 + d(i) P_{\text{route}}$ as a fallback.

Before running any outbreak we characterise each built network structurally, since topology bounds what an epidemic or an intervention can do. For every network we measure order and density, the degree distribution and its heterogeneity ratio $\langle k^2 \rangle / \langle k \rangle$ (which governs the metapopulation invasion threshold [15]), degree assortativity, average clustering, the mean shortest-path length and diameter, community modularity, and the rank correlation ρ between a city’s degree and its betweenness, together with the count of anomalous gateways, low-degree cities whose betweenness far exceeds that of their similar-degree peers. These gateways are the anomalous centrality that Guimerà et al. found in the worldwide air network [19]; we flag them not by an arbitrary cutoff but by standardising each city’s betweenness against that of its similar-degree peers and testing for a significant excess under Benjamini–Hochberg false-discovery-rate control, the degree-conditioned benchmark of [41].

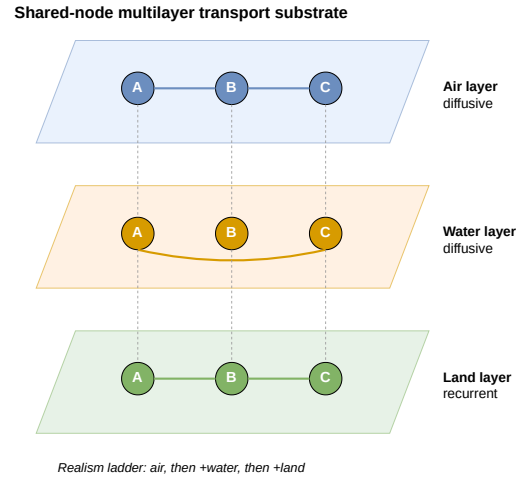


Figure 4: Shared-node multilayer substrate. Each city is one shared node present on every layer it uses (vertical dashed links); solid links are routes on that layer, drawn only where a route exists (not all-to-all). Air and water couple cities by diffusive relocation, land by recurrent commuting; the realism ladder adds water, then land.

2.2 Running an outbreak on the map

Each city runs a compartmental model internally, while individuals move between cities along the edges of G once per day. This is a metapopulation reaction–diffusion process [14]: a local reaction updates the compartments within each city, and a mobility step couples cities through the network. Within city i the force of infection is $\lambda_i = \beta I_i / N_i$, and a daily step applies the reaction to every city and then the mobility coupling before advancing to the next day. The mobility step is where the multilayer map earns its keep. Treating all travel as a single diffusive process is a known source of error [5], so the engine implements the two mechanisms matched to the physics of the layers, and we commit to this pair rather than collapsing them into one:

- Diffusive mobility (air, water). A fraction of each compartment leaves city i for neighbour j each day, proportional to the layer travel rate τ_ℓ and the normalised edge weight w_{ij} , and mixes into the destination population. This is the standard reaction–diffusion coupling and carries a global invasion threshold in mobility rate [15].
- Recurrent mobility (land, commuting). Commuters travel to a neighbour by day, mix there, and return home at night, so they are not relocated. We model this by coupling the force of infection rather than moving population: residents of i experience $\pi_i = \sum_j C_{ij} I_j^*/N_j^*$, where C is the per-capita commuting matrix (the land kernel with a stay-home diagonal) and I_j^*, N_j^* are the infectious and total people present at j . When no land layer is present C is the identity and $\pi_i = I_i/N_i$, recovering the plain diffusive model.

The distinction matters: diffusive travel relocates individuals, whereas recurrent commuting returns them home each day and so couples neighbouring cities far more tightly. The two layers we add to air therefore play opposite roles, water spreading the outbreak diffusively to new coastal cities while land synchronises whole regions through commuting. An optional extension lets transmission occur during travel (most relevant for long ferry crossings) but is disabled in the runs reported here.

2.3 Choosing what to spread

The engine runs whichever compartmental model it is given, so the choice of diseases is the substantive decision. Rather than commit to named pathogens, whose exact parameters are endlessly contestable, we define five disease types, each fixed by its dynamical structure and parameterised from one representative exemplar with values drawn entirely from the literature (Table 1). The five span the behaviours relevant to the question: an immunizing acute epidemic (measles-like, SIR), a latent one (COVID-like, SEIR), a lethal infection controllable by isolation (Ebola-like, SEIQRD), an endemic infection conferring no immunity (gonorrhea-like, SIS), and a recurrent one with waning immunity (influenza-like, SEIRS). Each type adds one mechanism relative to a simpler one, so a difference in outcome is attributable to that mechanism rather than to several at once. All five are run and reported; where a single contrast makes a point most sharply we lead with the two extremes, the immunizing SIR and the endemic SIS, and read the three intermediate types against them.

2.4 Choosing how to defend

With the disease types fixed, the remaining variable is the intervention, and each intervention is a decision about which structural elements of the network to remove. We compare two modes. The first is node targeting (vaccination): a set of cities is rendered immune so they no longer transmit. The targeting rule is the object of study, and at a fixed budget we compare no intervention (control), random selection, degree (the highest-traffic cities), betweenness (the structural bridges), a motif rule we label subgraph [17; 31], which ranks cities by their embedding in dense local clusters (triangle participation, the number of closed triads through each

node) rather than by raw connectivity, and two optimal-percolation rules from the modern immunization literature: Collective Influence [32], which scores a city by the summed reduced degrees of the nodes on a ball of radius two around it, and a non-backtracking rule [38], which ranks cities by their weight in the leading eigenvector of the Hashimoto (non-backtracking) matrix. Subgraph centrality weights a node by its participation in closed walks, so it is high precisely where a city sits in a cohesive, triangle-rich cluster in which infection can recirculate over redundant paths rather than pass through once [17]. This is the structural signature of local transmission reinforcement: where degree counts a city’s routes and betweenness the paths it bridges, subgraph centrality identifies the tightly interconnected neighbourhoods that sustain an outbreak once it arrives, and these need not be the busiest or the most central cities. It therefore probes a mechanism of spread the other rules miss, which is why we include it in the comparison. The two optimal-percolation rules are designed to reward exactly the low-degree, high-impact cities of the anomalous-gateway kind rather than raw hubs, and so test whether a method built for uncorrelated networks gains anything on a correlated, real one. Having found the most effective rule, we then sweep its budget, from a handful of cities to a few hundred, to trace the dose–response: how many cities must be vaccinated for substantial containment, and where the returns diminish. Coverage and per-city efficacy are held fixed throughout, so only the budget and the rule vary.

The second mode is edge targeting (interdiction): we close routes rather than immunise cities, in four scenarios: (A) no intervention; (B) close all air routes while retaining land and water; (C) close all air routes within an air-only model; and (D) close only the top- k airports (removing all of their routes) ranked by degree versus by betweenness. Scenarios B and C exist to contrast what a single-layer model concludes with how the full multilayer network behaves. The two modes are distinct levers, not stacked, so each reads in the same currency: containment per unit of a fixed budget.

2.5 Parameters and protocol

Each disease type fixes its clinical rates from the literature (Appendix A, Table 3): recovery γ , and where present incubation σ , isolation κ , and waning ω , each the reciprocal of a measured duration. We then set the transmission rate β so the within-city reproduction number matches that type’s literature R_0 . Because each city is well mixed internally, that local number is β/γ for SIR, SIS, and SEIR, and $\beta/(\gamma + \kappa)$ for the isolation type, where isolation shortens the infectious period. For the two extremes this gives measles $\gamma = 0.13$, $R_0 \approx 15$, hence $\beta = 1.95$, and gonorrhea a long infectious duration $\gamma \approx 0.03$ with $R_0 \approx 1.4$, hence $\beta \approx 0.042$; the remaining three (COVID, Ebola, influenza) follow the same rule. We deliberately apply no $\langle k^2 \rangle / \langle k \rangle$ correction: that adjustment is for networks whose nodes are individuals, whereas here nodes are well-mixed cities. Every rate and R_0 traces to a specific study, listed with its source in Appendix A.

We do not calibrate to outbreak data, since the transport network carries no ground-truth epidemic to fit, so absolute incidence is illustrative rather than predictive. Every run is seeded inside the largest connected component: the sparser layers (ferries especially) bring in coastal cities that sit in small disconnected pockets, where a seed cannot reach the mainland and the outbreak fizzles, and mixing those fizzles

Table 1: The disease types studied. Each type is defined by its dynamical structure (its compartmental model) and parameterised from one representative exemplar disease. Numeric parameter values and their literature sources are given in Appendix A (Tables 3 and 4).

#	Type	Model	Example diseases	Parameters from
1	Immunizing, acute	SIR	measles, rubella, mumps	measles
2	Latent + immunizing	SEIR	COVID-19, SARS, chickenpox	COVID-19
3	Lethal, isolation-controlled	SEIQRD	Ebola, Marburg, pneumonic plague	Ebola
4	Endemic, no immunity	SIS	gonorrhea, chlamydia, common cold	gonorrhea
5	Recurrent, waning immunity	SEIRS	seasonal influenza, RSV, norovirus	seasonal influenza

with full epidemics would conflate two questions, whether an outbreak takes off and how large it grows. We therefore seed where it can spread, and measure the second. A single uniform horizon of 1460 days (four years) gives cross-disease comparability: the acute type peaks and burns out within months, while the endemic type needs years to approach its equilibrium. Across any comparison the only inputs that change are the disease type, the protection strategy, and the layer set. We report peak active infection (as a fraction of the population), final epidemic size, and peak day; for the lethal isolation type we additionally report cumulative deaths and deaths averted per capita, read from its fatal compartment, and for each targeting rule the geographic concentration of the cities it selects, summarised by a per-country Gini coefficient. To quantify seed sensitivity, every configuration is run as an ensemble of ten random seedings of the initial infection, and each headline number is the ensemble mean with a 95% confidence interval ($\text{mean} \pm 1.96 \text{ s.e.}$), so every reported difference is resolved against seed variation. We verify the engine against analytic limits, namely single-node dynamics and the final-size correspondence between SIR and bond percolation [33].

These pieces form a single staged protocol (Figure 5), walked one factor at a time rather than as a full grid. We first run each disease with no intervention to see how it spreads; then defend, comparing every targeting rule at a fixed budget per disease and keeping each one’s best, so “which protection wins” is read off per disease rather than assumed from one; then trace the dose–response by sweeping each winner’s budget. The same protocol evaluates the edge-targeting lever, interdiction, progressively closing routes (air, then ferries) on the flagship. The vaccination phases run up the realism ladder (air, then +water, then +land), so each result reads against both the disease and the realism axis. In all, the staged protocol comprises roughly 1,600 independent simulations across the eight networks, each one run of the engine at a fixed configuration and seed; walking the factors sequentially keeps this well below the full Cartesian grid the same axes would otherwise generate.

2.6 Implementation and reproducibility

The study is a single Python codebase (NumPy and NetworkX) organised as three modules, data retrieval, network generation, and epidemic evaluation, over one shared engine (Appendix C). The interface is an interactive web simulator that configures an outbreak and streams its progression as live epidemic curves and geographic maps; the same engine and configuration are also driven programmatically to run the staged protocol over many configurations. Either way the engine, configuration, and output are identical, so results do not depend on how a simulation is launched, and

every run is fully determined by its configuration and seed.

3. RESULTS

We read every outcome in one currency, the peak of active infection, and walk the staged protocol of Section 2.5 one factor at a time: how each disease spreads with no intervention, which node-targeting rule best defends, how that winner scales with dose, and what closing routes (interdiction) achieves. All runs are on Europe, with the disease type and the realism ladder (air $\rightarrow$ +water $\rightarrow$ +land) as the two cross-cutting axes. Every configuration is run as an ensemble of ten random seedings of the initial infection; we report the mean and a 95% confidence interval across the ensemble, so absolute counts are comparative rather than predictive and every difference we read below is resolved against seed variation.

3.1 The transport networks

Structure sets what any epidemic or countermeasure can do, so we characterise the substrates before intervening (Table 2; the full descriptor set is Table 5). All are sparse (density ≤ 0.033) yet strongly small-world: the mean shortest path is only about three hops ($\langle \ell \rangle = 2.7$ on Europe air, 3.2 on the multilayer) while local clustering stays high ($C \approx 0.5$), so any two cities lie a few steps apart and an outbreak can cross the continent in weeks. The degree distribution is heavy-tailed across two orders of magnitude (Figure 12, Appendix B), with a heterogeneity ratio $\langle k^2 \rangle / \langle k \rangle$ of 91–121; in a metapopulation this heterogeneity is what lowers the invasion threshold and gives the busiest airports their outsized seeding role [15].

Climbing the realism ladder reorganises that structure without changing its small-world character. Adding land and ferry layers roughly triples the modelled population and flips the degree assortativity from disassortative on air alone ($r = -0.19$) to mildly assortative on the multilayer ($r = +0.14$), as commuting joins cities of similar size; it stretches the diameter from 6 to 11 and lifts modularity from $Q = 0.28$ to 0.35 as national road and ferry clusters form.

Most consequential is how a city’s degree relates to its betweenness. In most cities the two rank together, so the busiest airports are also the main bridges and it scarcely matters which a vaccination rule targets. The exceptions are anomalous gateways: cities that lie on a great many shortest paths (high betweenness) despite having only a handful of direct routes (low degree), so they carry an epidemic across the continent far out of proportion to their traffic. This is the anomalous centrality Guimerà et al. found in the worldwide air network, with Anchorage, few flights but the gateway to remote Alaska, the canonical example [19]. How many a network has is a property of the region: the correla-

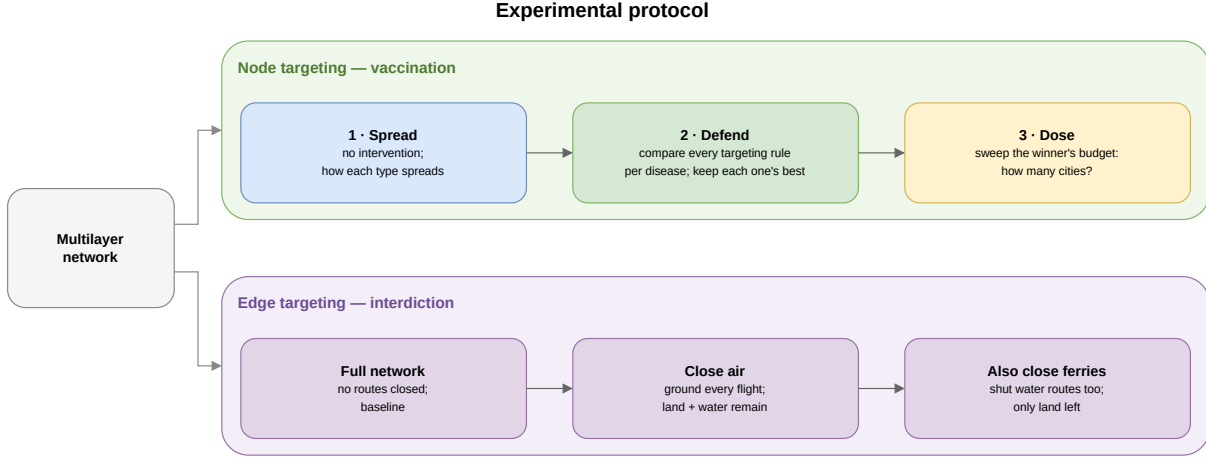


Figure 5: Experimental protocol. From one multilayer network the study splits into two protection levers. Node targeting (vaccination) is walked one factor at a time: spread, then defend (compare every targeting rule per disease and keep each one’s best), then sweep the winner’s budget (dose). Edge targeting (interdiction) progressively closes routes (air, then ferries). Both levers are run for all five disease types (led by the immunizing SIR and endemic SIS extremes), the vaccination phases each up the realism ladder air $\rightarrow$ +water $\rightarrow$ +land.

tion ρ (degree, betweenness) spans a spectrum from a US-like regime where hubs and bridges coincide and anomalies all but vanish [41] to a worldwide-like regime rich in gateways. Europe’s air network is strongly correlated ($\rho = 0.89$, US-like), but adding the land and ferry layers pulls it toward the anomalous end ($\rho = 0.74$) and surfaces seventeen gateways on the flagship, low-degree bridge cities on the Nordic and western periphery (Figure 6; the degree–betweenness scatter is Figure 13 in Appendix B). These gateways are the only cities where degree- and betweenness-targeting disagree, which makes them the whole stake of the defence comparison below: if targeting bridges beats targeting hubs, it can only be because it protects them.

3.2 How each disease spreads

With no intervention every one of the five types takes off on every rung of the ladder (Figure 7a). We report the peak as a fraction of the population, so that substrates carrying different numbers of people are compared fairly. The peak then orders cleanly by dynamical class: the immunizing measles type (SIR) is highest, at 59% of the population (95% CI ± 1.4) on air alone, followed by the endemic gonorrhea type (SIS, 28%) and the latent COVID type (SEIR, 12%), with the waning-immunity influenza type (SEIRS, 4.0%) and the isolation-controlled Ebola type (SEIQRD, 3.8%) lowest. The intervals are tight (at most ± 2 points on the measles type and under half a point on the milder ones), so the ordering is resolved across seeds. Latency delays a peak and isolation caps it, so the two types that add those mechanisms surge least, even though every type clears its invasion threshold [15].

Climbing the realism ladder moves that fraction only a little, and chiefly for the measles type (air 59%, +water 57%, +land 63%); the milder types stay within half a point, which is why we normalise. In absolute counts the peak roughly doubles from air to the full multilayer (measles 117M $\rightarrow$ 216M; gonorrhea 56M $\rightarrow$ 97M), but that rise is almost entirely a population effect: the land layer pulls inland, non-airport cities into the network, growing the modelled

population from 199M to 342M. Per capita, the ladder adds people who can be infected, not severity. Ferries even lower the peak slightly (measles 59% $\rightarrow$ 57%), because the coastal cities they add crest out of phase and desynchronise the surge, whereas land’s role is dynamical, synchronising a larger connected population through recurrent commuting [5] and bringing the peak forward (Figure 7b). The temporal signature, by contrast, belongs to the disease and not the substrate: the acute measles type peaks within about three weeks (day 17–21) and burns out, while the endemic gonorrhea type is still climbing toward its plateau at the four-year horizon.

3.3 Which vaccination wins

Holding the budget at 15 immunised cities (75% coverage) on the multilayer flagship, targeting the structural bridges (betweenness) yields the largest peak reduction for every one of the five diseases (Figure 8): measles 12.5%, the latent, waning, and isolation-controlled types 12.6–12.9%, and the endemic gonorrhea type 10.8%. Targeting the busiest hubs (degree) and the dense-cluster subgraph rule (Section 2.4) tie well below it (7.0–9.3%), while random immunisation is barely distinguishable from doing nothing ($\approx 1\%$). The confidence intervals are a few tenths of a point, so the ordering bridges $>$ hubs $\approx$ subgraph $\gg$ random is unambiguous; it holds for all five disease types and, in the underlying table, at all three rungs, so the win is not an artefact of one disease or one substrate [25]. And the gateways are what decide it. Because degree and betweenness diverge only at those cities (Section 3.1), betweenness’s roughly four-point edge over degree is precisely the containment the seventeen low-degree bridges carry, which degree-targeting forgoes by ranking them too low to protect.

The two optimal-percolation rules put that mechanism to a further test: does a method beat degree only if it can see past raw connectivity to the gateways? They split cleanly. Collective Influence [32], which scores a city by its neighbourhood’s reach rather than its own traffic, nearly matches betweenness (measles 11.8% against 12.4%, and within half

Table 2: Key structural descriptors per network (topological, no simulation). $\langle k \rangle$ mean degree; $\langle k^2 \rangle / \langle k \rangle$ the degree heterogeneity governing the invasion threshold; r degree assortativity; $\langle \ell \rangle$ mean shortest-path length (small-world); Q community modularity; ρ Spearman correlation of degree and betweenness with the number of anomalous gateways. Only the Europe substrates are shown here, since the main text focuses on Europe; the full descriptor set for every built network is Table 5 in Appendix B.

Network	n	$\langle k \rangle$	$\langle k^2 \rangle / \langle k \rangle$	r	$\langle \ell \rangle$	Q	ρ	#gw
europe/air	524	34.2	121.0	-0.193	2.70	0.277	0.894	7
europe/air+land	1234	27.2	92.5	0.125	3.00	0.318	0.744	17
europe/air+land+water	1359	25.5	91.3	0.138	3.15	0.345	0.741	17
europe/air+water	674	28.3	116.7	-0.121	3.10	0.306	0.850	16

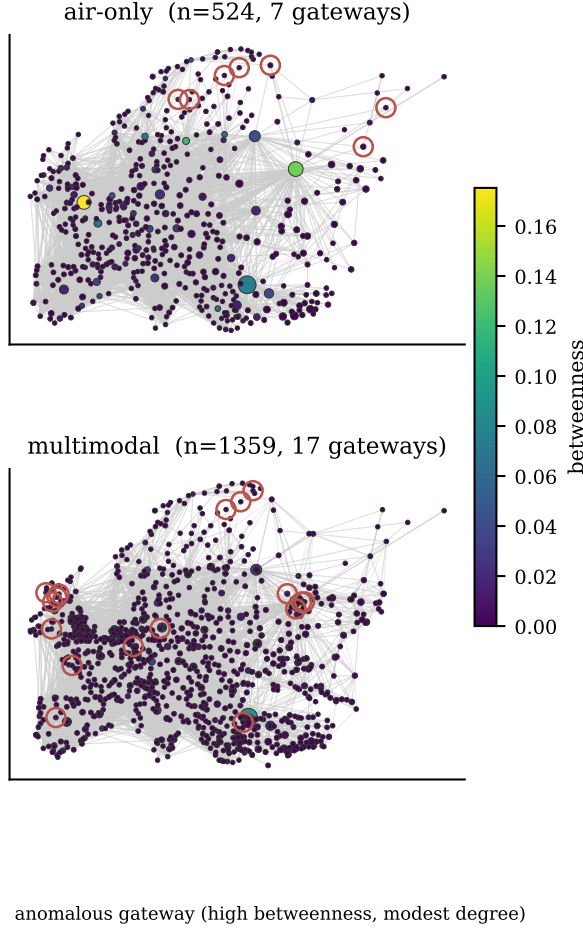


Figure 6: The Europe network on real geography: air-only (top) versus multimodal (bottom). Node colour is betweenness, size is population, and rings mark anomalous gateways. Adding land and ferry layers redistributes betweenness onto peripheral bridge cities (Section 3.1).

a point on every disease), recovering most of the advantage that degree forfeits at the gateways. The non-backtracking rule [38], by contrast, tracks degree (7.8–9.2%): its eigenvector ranking rewards the same busy cities degree already finds and so misses the same gateways. The two rules that beat degree are exactly the two that can find the low-degree bridges, and the one that cannot forfeits them alongside degree, so the test set up in Section 3.1 resolves squarely in

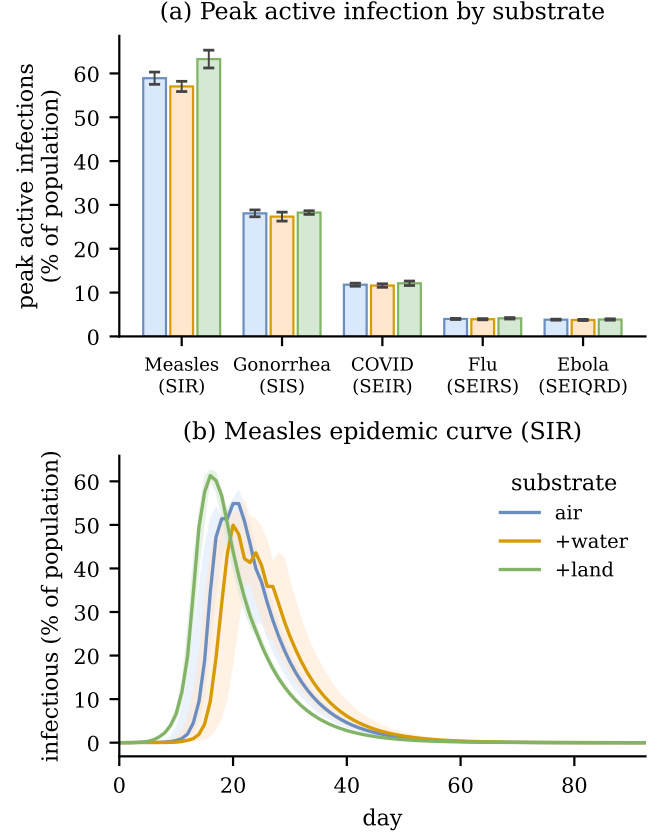


Figure 7: How each disease spreads, with no intervention. (a) Peak active infection, as a fraction of the population, by disease type and substrate up the realism ladder (air, +water, +land); normalising by population removes the confound that the land layer also adds inland cities. The peak orders by dynamical class (immunizing > endemic > latent > waning $\approx$ isolation-controlled) and is nearly flat across the ladder: realism adds people, not per-capita severity. (b) Epidemic curve of the immunizing measles type across the same ladder: infectious fraction over time, seed median with an interquartile band. Adding land brings the peak earlier and higher (recurrent commuting synchronises a larger population), while ferries push it slightly later and lower (coastal cities crest out of phase), so the substrate reshapes the timing even where it barely moves the per-capita height. Error bars in (a) are 95% confidence intervals and the band in (b) an interquartile range, both across the ten-seed ensemble (Section 3.2).

the gateways’ favour. Betweenness remains the rule to use, with Collective Influence a confirming second.

The bridges-over-hubs ordering holds at every rung, but vaccination gets harder as the substrate grows more realistic. Immunising 15 cities cuts the measles peak by 19% under betweenness on the sparse air-only network, yet only 12% on the full multilayer, with degree falling in step (12% $\rightarrow$ 8%). Two effects compound: the fixed 15-city budget covers a smaller share of the larger multilayer, and its land and ferry layers give the outbreak redundant routes around any removed city. Betweenness keeps its $\approx 50\%$ relative edge over degree throughout, so realism does not change which cities to protect, only how much budget a given reduction demands, which is the same redundancy that blunts route closure below.

3.4 How much budget it takes

Sweeping the winning betweenness budget from 5 to 200 cities on the flagship traces a diminishing-returns curve for every disease (Figure 9): peak reduction climbs from $\approx 8\%$ at 5 cities to 30–34% at 200, passes $\approx 12\text{--}13\%$ at the 15-city operating point used above, and flattens beyond $\sim 60\text{--}120$ cities. Even the best rule, then, needs a substantial budget to make a large dent, and no small set of cities contains these outbreaks. The disease ordering is consistent at every budget: the types that confer immunity (SIR, SEIR, SEIQRD, SEIRS) respond slightly better than the endemic SIS (at 200 cities, Ebola 33.8% down to gonorrhea 30.1%), because a recovered compartment locks in the vaccination’s gain whereas SIS reinfection steadily erodes it.

3.5 Closing routes instead of cities

The second lever closes routes rather than immunising cities, and it is where a single-layer model and the full multilayer part company most sharply (Figure 10). Closing all air routes in an air-only model collapses every outbreak to 0.1–0.2% of its full peak (measles 0.18%, gonorrhea 0.12%): grounding flights ends the epidemic when air is the only carrier. On the true multilayer substrate the same closure still leaves 32–47% of the peak standing (measles 47%, COVID 33%, Ebola 32%), because land commuting and ferries keep moving people. A single-layer picture therefore overstates the benefit of a flight ban by more than two orders of magnitude.

Closing only the top-10 hub airports (ranked by betweenness) is weaker still: it leaves 61–71% of the peak in the air-only model and 77–90% on the multilayer, and for the slow endemic type nearly all of it (90%). Removing a handful of bridge airports barely perturbs an outbreak that also travels by road and sea. Either way the message is the same, that the realistic multimodal substrate flips the policy conclusion a single-layer model would reach: once ground and ferry travel are in the picture, no air-route closure contains the epidemic on its own.

3.6 Deaths averted, and where protection lands

Two refinements sharpen the vaccination result. First, peak active infection understates what is at stake for the lethal isolation type, whose case-fatality is high ($\mu \approx 0.71$). Reading cumulative deaths from the model’s fatal compartment, bridge-targeting averts $\approx 7,400$ deaths per 100,000 people on the flagship, against $\approx 4,800$ for hub- and subgraph-targeting

and only ≈ 680 for random selection (Figure 11a). The bridge advantage is therefore larger under a mortality metric (53% more deaths averted than degree) than under the peak metric ($\sim 39\%$), because the bottleneck cities betweenness protects sit on the very routes that would otherwise carry the disease into spared populations.

Second, effectiveness and equity need not trade off on this substrate. Betweenness-targeting places its cities across the continent rather than onto one nation’s hubs: the operating top 15 span 12 countries, and larger budgets (50, 200) extend that cover outward (Figure 11b). Degree, the subgraph rule, and both optimal-percolation strategies instead concentrate their 15 cities in 10 countries with 27% in a single one (Germany), a Gini of 0.26 against betweenness’s 0.15. The most effective rule is thus also the most geographically even: targeting structural bridges distributes protection across the continent’s borders rather than piling it onto one nation’s hubs.

4. DISCUSSION

The results answer the paper’s question in a way that depends on how realistic the substrate is, and they do so consistently across all five disease types. Three threads run through them: how the disease type shapes the epidemic, why targeting bridges beats targeting hubs, and why the choice of substrate changes what containment is even possible.

4.1 Disease type shapes the curve

The five types separate cleanly by their dynamical structure rather than by any single rate. The immunizing measles type produces the tall, right-skewed, single-peaked curve characteristic of SIR: infectious cities find abundant susceptible neighbours, incidence grows, and then the susceptible pool is depleted irreversibly, so the outbreak burns out [24]. The endemic gonorrhea type has no recovered compartment, so reinfection continually replenishes susceptibles and incidence settles at an endemic equilibrium rather than declining to zero. The three intermediate types interpolate as expected: latency (SEIR) delays and lowers the peak, and both isolation (SEIQRD) and short infectious windows blunt it.

One coincidence deserves care. The waning-immunity influenza type and the isolation-controlled Ebola type land at almost the same peak height ($\sim 8\text{M}$ on air), but for mechanistic rather than biological reasons: influenza’s basic reproduction number is low ($R_0 \approx 1.5$), while Ebola’s is higher ($R_0 \approx 2$) but its isolation shortens the effective infectious period ($1/(\gamma + \kappa) \approx 3.3$ days), so the peak, which depends on the infectious period as well as R_0 , settles at a similar level. This tells us the model is internally coherent, not that an Ebola outbreak would resemble a flu season. It would not: the isolation type is exactly where the well-mixed metapopulation framing is weakest (Section 4.4), so its absolute number is the least literal of the five, and it is run on the same footing as the others only for the sake of comparison.

4.2 Why bridges beat hubs

Across every disease and every rung, immunising the highest-betweenness cities reduced the peak more than immunising the highest-degree cities, and both beat random selection. The mechanism is structural. Because transmission on a network follows paths, and betweenness counts how many shortest paths run through a node, removing high-

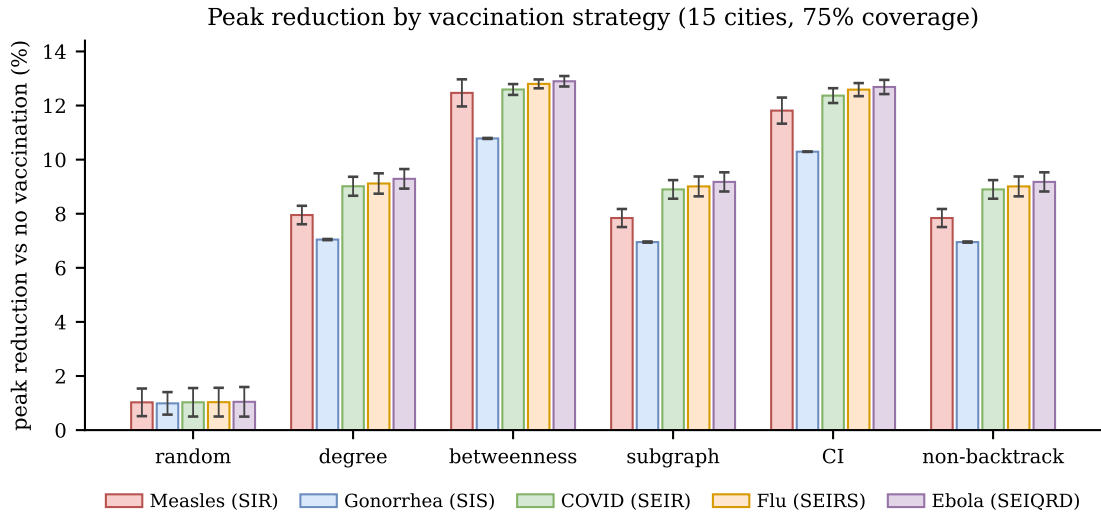


Figure 8: Node targeting. Peak reduction versus no vaccination for each strategy at a fixed budget (15 cities, 75% coverage) on the multilayer flagship. Betweenness (bridges) wins for every disease type, with Collective Influence a close second; degree (hubs), the subgraph rule, and non-backtracking trail together, and random is near zero. Error bars are 95% confidence intervals across the ten-seed ensemble; they are a few tenths of a point and often smaller than the bar borders (Section 3.3).

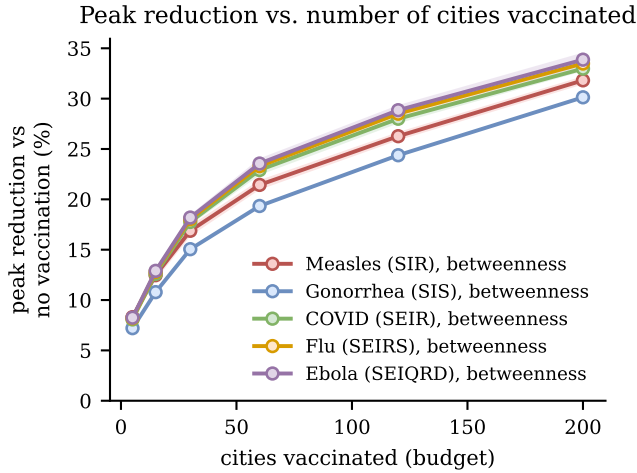


Figure 9: Dose-response. Peak reduction versus the number of cities vaccinated under the winning betweenness rule, on the flagship. Returns diminish beyond ~ 60 – 120 cities; immunising diseases gain slightly more than the reinfecting endemic type. Shaded bands are 95% confidence intervals across the ten-seed ensemble; they are very narrow (often thinner than the marker), reflecting the low seed-to-seed spread in peak reduction (Section 3.4).

betweenness cities severs the routes that connect otherwise weakly coupled regions [25]. Removing high-degree cities instead deletes locally busy nodes whose traffic is often redundant, since their neighbours can still reach each other by other short paths, so the giant component and its shortest-path structure survive more intact. Betweenness targets the bottlenecks of reachability; degree targets a volume the network can partly route around. Concretely, the difference lives in the anomalous gateways: the low-degree, high-betweenness bridge cities that a degree ranking places too

low to protect, even though a disproportionate share of cross-continental transmission must pass through them [19]. Degree and betweenness agree everywhere else, so these gateways are the whole of betweenness’s margin, and the result is really a statement about them: the rules that win are the ones that find the bridges that traffic alone hides.

The realism ladder does not change this ordering, but it raises the stakes. Immunising the same 15 cities cuts the peak by about 19% on the air-only network and only 12% on the full multilayer, with degree falling in step (12% to 8%): the fixed budget covers a smaller share of the larger network, and its land and ferry layers give the outbreak redundant paths around any removed city, so every rule loses absolute grip while betweenness keeps its roughly 50% relative advantage over degree [40]. The multilayer substrate thus makes node targeting harder, not easier, through the very redundancy that also lets outbreaks survive route closure (Section 3.5); a single-layer model, seeing only air routes, would overstate how much any vaccination campaign achieves.

4.3 Realism mostly adds people, and changes what containment means

Climbing the ladder raised the absolute peaks, but the spread results (Section 3.2) show that most of that rise is a population effect: the land layer brings inland, non-airport cities into the network, and the peak, as a fraction of the population, stays nearly flat across rungs. This is why we report that fraction rather than the raw count, which would let the land layer’s extra population masquerade as extra severity. Ferries change little, because they reach only a handful of coastal cities; land’s real contribution is dynamical, synchronising a larger connected population through recurrent commuting [5].

The interdiction results carry the sharpest practical implication. In an air-only model, grounding all flights ends the outbreak; on the multimodal substrate the same closure leaves a third to a half of the peak standing, and closing only the busiest hubs does almost nothing. Whether route closure can

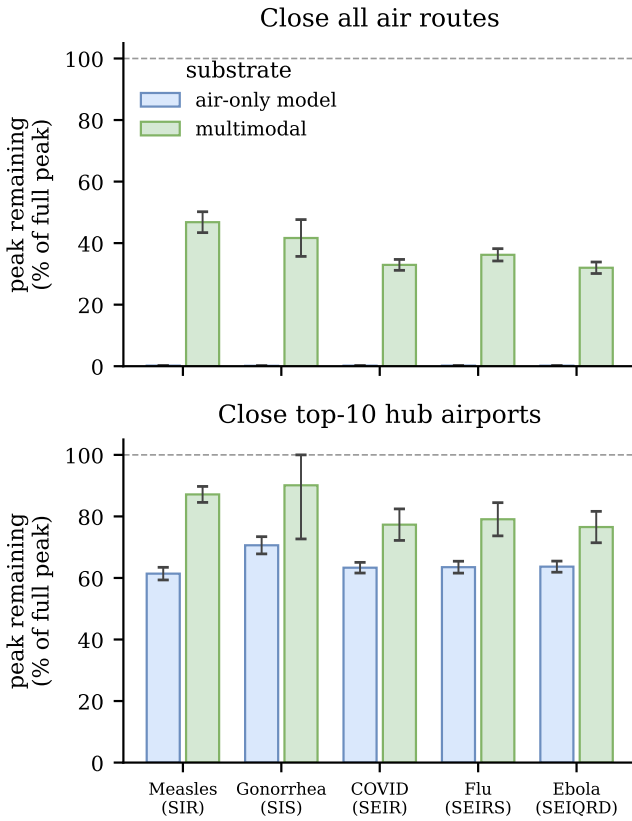


Figure 10: Edge targeting. Peak remaining (as a percentage of each substrate’s own full-network peak) after closing all air routes (top) or the top-10 hub airports (bottom), for the air-only model versus the multimodal network. Closing air ends the outbreak only in the air-only world (its bars sit near zero); on the multimodal substrate 32–47% of the peak survives. Error bars are 95% confidence intervals across the ten-seed ensemble, with the upper whisker capped at 100% since peak remaining cannot exceed the full-network peak (Section 3.5).

contain an epidemic at all thus flips with the substrate. This echoes the broader lesson that single-layer transport models systematically mislead about containment once ground and maritime mobility are included [4], and it is the clearest argument in the paper for building the multilayer substrate rather than the air network alone.

4.4 Limitations

The study is comparative, not predictive, and its absolute counts should not be read as forecasts. Three limitations bound the interpretation. First, the well-mixed metapopulation makes transmission proportional to the local infectious fraction, which fits casual-contact respiratory and endemic diseases far better than a contact-limited pathogen such as Ebola: real filovirus transmission requires direct contact with symptomatic or deceased cases and is shaped by care settings that a constant isolation rate only crudely approximates. Its peak is therefore the least trustworthy of the five, and our metric, peak active infection, does not even reflect the isolation type’s high case-fatality ($\mu \approx 0.71$); we therefore also report deaths averted per capita for the lethal type

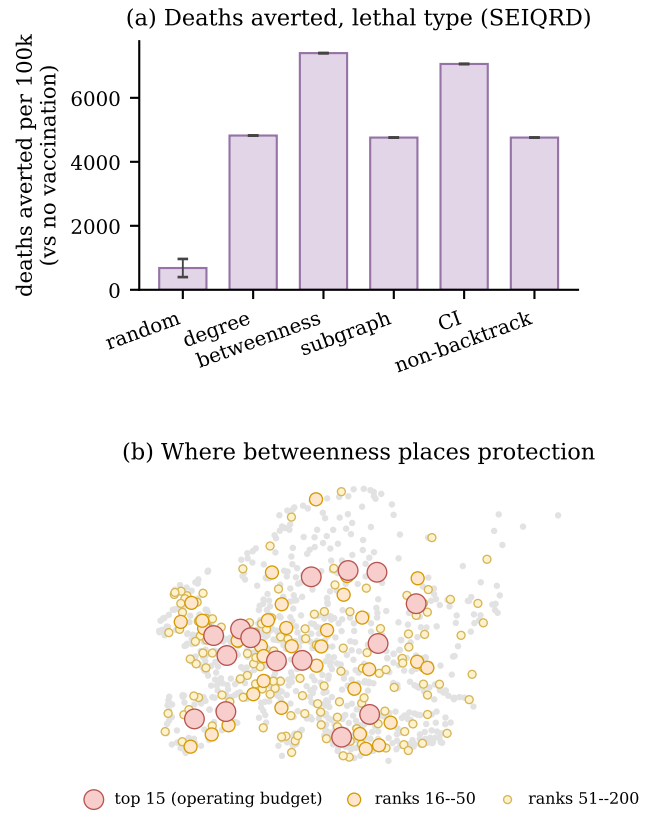


Figure 11: Deaths averted, and where protection lands. (a) Deaths averted per 100,000 people for the lethal isolation type (SEIQRD) under each strategy at the 15-city operating point; bridge-targeting averts the most, and by a wider margin than under the peak metric. Error bars are 95% confidence intervals across the ten-seed ensemble. (b) Where betweenness-targeting places protection on the flagship, at the three dose budgets. Cities sit at their coordinates (grey: all cities); the operating top 15 (large red) spread across 12 countries rather than clustering in one, and the next tiers (ranks 16–50, 51–200) extend that cover outward across the continent (Section 3.6).

(Section 3.6), where bridge-targeting shows its largest margin. Second, every configuration is run as a ten-seed ensemble and we attach a 95% confidence interval to each headline number, so the reductions carry uncertainty bounds and the strategy ordering is resolved against seed variation; a finer coverage grid and full transition-level stochasticity remain natural extensions. Third, parameters are justified from the literature rather than calibrated to any outbreak, and the transport weights are proxies for passenger volume, so the results are internally consistent comparisons, not fits to data. These bounds are also the roadmap. The same engine runs stochastic ensembles, other regions, and finer coverage grids without modification, and the interactive simulator turns each of those into a configuration change rather than a code change, so the findings here, bridges over hubs, realism amplifying that gap, and the limits of route closure, form a stable core on which a more quantitative, region-specific study can build.

5. CONCLUSION

We compared how five disease types spread across a European multilayer transport network, and which network-science interventions best contain them, holding every factor but the one under study fixed. Three findings recur across all five types and are resolved against a ten-seed ensemble. First, the disease type orders the outbreak by dynamical class rather than by any single rate, while climbing the air $\rightarrow$ +water $\rightarrow$ +land realism ladder mostly adds people rather than per-capita severity: the peak fraction stays nearly flat, and ferries even desynchronise it. Second, immunising the structural bridges (betweenness) beats immunising the busiest hubs (degree) for every disease, and it is also the most geographically even choice and the one with the widest margin once mortality rather than peak prevalence is the yardstick. Third, the multilayer substrate changes what containment means: it makes node targeting harder, since land and ferry redundancy routes the outbreak around any removed city, and it overturns the interdiction conclusion a single-layer model would reach, where grounding flights ends an air-only epidemic but leaves a third to a half of the multimodal one standing.

Underlying these results is the network structure itself. The transport substrates are sparse but strongly small-world, with heavy-tailed degrees and a community organisation that the land and ferry layers sharpen, and their added bridges surface as low-degree, high-betweenness anomalous gateways: precisely the cities betweenness targeting protects and degree targeting misses. The study is comparative rather than predictive, and its absolute counts are not forecasts. The same engine extends without modification to other regions, finer coverage grids, calibrated parameters, and full transition-level stochasticity, so the stable core established here, bridges over hubs, realism raising the cost of containment, and the limits of route closure, is a foundation on which a more quantitative, region-specific study can build.

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APPENDIX

A. DISEASE PARAMETERS

Every parameter value is taken from the literature, not tuned. Each clinical rate is the reciprocal of a measured duration of the exemplar disease (rate = 1/duration); each basic reproduction number R_0 comes from a separate epidemiological study; and the transmission rate β is derived from both, $\beta = R_0\gamma$ (or $\beta = R_0(\gamma + \kappa)$ for the isolation type). Table 3 lists the value of every rate per type, and Table 4 gives the source for each. Notably, measles and Ebola have similar infectious periods (~ 8 – 10 d) yet R_0 differs roughly sevenfold (≈ 15 vs ≈ 2): the infectious period does not set R_0 ; together with β it produces it.

Table 3: Parameter values per disease type (rates per day; rate = 1/duration, with the duration in parentheses). No $\langle k^2 \rangle / \langle k \rangle$ correction is applied, as nodes are well-mixed cities rather than individuals. The endemic type’s γ and R_0 are illustrative: with no immunity and a long, variable untreated duration, the regime matters, not the exact number. For the lethal type $\gamma_Q \approx \gamma \approx 0.10$.

#	Type · model	γ	σ	κ	ω	μ	R_0
1	Immunizing · SIR	0.13 (~ 8 d)	—	0	0	≈ 0	12–18
2	Latent · SEIR	0.13 (~ 7 – 8 d)	0.20 (~ 5.1 d)	0	0	~ 0.01	2–3.3
3	Lethal · SEIQRD	0.10 (~ 10 d)	0.09 (~ 11.4 d)	0.20 (~ 5 d)	0	0.71	1.5–2.5
4	Endemic · SIS	~ 0.01 – 0.05	—	0	∞	0	low ($\gtrsim 1$)
5	Recurrent · SEIRS	0.21 (~ 4.8 d)	0.70 (~ 1.4 d)	0	~ 0.001 (2–7 yr)	~ 0.001	1.3–1.8

Table 4: Source for every disease-type parameter. γ , σ , κ come from a measured clinical duration of the exemplar; R_0 from a separate epidemiological study; β is derived from both.

Disease (model)	Value	Measured quantity	Source (paraphrased)
Measles (SIR)	$\gamma = 0.13$ $R_0 = 12$ – 18 $\omega = 0$	infectious ~ 8 d R_0 immunity	contagious 4 d before to 4 d after rash [12] “often cited to be 12–18” (review of 58 estimates) [18] natural infection confers life-long immunity [12]
COVID (SEIR)	$\sigma = 0.20$ $\gamma = 0.13$ $R_0 = 2$ – 3.3	incubation 5.1 d infectious ~ 1 wk R_0	“mean incubation period 5.1 days (95% CI 4.5–5.8)” [27] infectiousness peaks near onset, low by ~ 7 d [21] “pooled R_0 3.32 (95% CI 2.81–3.82)” [1]
Ebola (SEIQRD)	$\sigma = 0.09$ γ, κ struct. $\mu = 0.71$ $R_0 = 1.5$ – 2.5	incubation 11.4 d onset $\rightarrow$ hospital ~ 5 d case-fatality R_0 (SEIR fit)	“incubation period was 11.4 days” [43] SEIHFR; community / hospital / burial transmission [28] “case fatality rate 70.8% (95% CI 69–73)” [43] 1.51 Guinea, 2.53 Sierra Leone, 1.59 Liberia [2]
Gonorrhea (SIS)	SIS, $\omega = \infty$ $\gamma \approx 0.03$	immunity, latency untreated duration	“negligible protective immunity, negligible latent period” [23] median untreated pharyngeal gonorrhea ≈ 16 weeks [6]
Influenza (SEIRS)	$\sigma = 0.70$ $\gamma = 0.21$ $\omega \approx 0.001$ $R_0 = 1.3$ – 1.8	incubation 1.4 d shedding 4.8 d immunity R_0 (review)	“median incubation period for influenza A 1.4 days” [29] “viral shedding averaged 4.80 days (95% CI 4.31–5.29)” [11] influenza A immunity wanes in 2–7 yr via antigenic drift [44] median $R \approx 1.80$ for the 1918 pandemic (IQR 1.47–2.27) [8]

B. FULL NETWORK DESCRIPTORS

Table 5 gives the complete structural descriptor set for every built network, extending the key-descriptor summary in the main text (Table 2). All quantities are topological, computed from the built graphs with no simulation; path length, diameter, clustering, and modularity are evaluated on the undirected largest connected component.

Table 5: Full structural descriptor set per network (topological). n/m nodes/edges; δ density; $\langle k \rangle$ mean degree; $\langle k^2 \rangle / \langle k \rangle$ heterogeneity; r assortativity; C average clustering; $\langle \ell \rangle / d$ mean shortest-path length and diameter (undirected largest component); GC giant-component fraction; Q Louvain modularity; λ_{NB} leading non-backtracking eigenvalue and $T_c = 1/\lambda_{NB}$ the structural percolation threshold; ρ degree-betweenness correlation; #gw anomalous gateways.

Network	n	m	δ	$\langle k \rangle$	$\langle k^2 \rangle / \langle k \rangle$	r	C	$\langle \ell \rangle$	d	GC	Q	λ_{NB}	T_c	ρ	#gw
europe/air	524	8972	0.0327	34.2	121.0	-0.193	0.499	2.70	6	1.000	0.277	54.86	0.0182	0.894	7
europe/air+land	1234	16788	0.0110	27.2	92.5	0.125	0.581	3.00	6	1.000	0.318	56.74	0.0176	0.744	17
europe/air+land+water	1359	17326	0.0094	25.5	91.3	0.138	0.541	3.15	11	0.985	0.345	56.81	0.0176	0.741	17
europe/air+water	674	9532	0.0210	28.3	116.7	-0.121	0.415	3.10	11	0.958	0.306	54.92	0.0182	0.850	16
africa/air	249	1188	0.0192	9.5	32.4	-0.225	0.511	3.19	7	1.000	0.607	12.97	0.0771	0.766	4
americas/air	986	8875	0.0091	18.0	111.3	-0.235	0.494	3.32	10	0.987	0.439	47.63	0.0210	0.760	18
asia/air	766	7689	0.0131	20.1	87.5	-0.157	0.544	3.14	7	1.000	0.529	41.09	0.0243	0.838	8
oceania/air	207	883	0.0207	8.5	31.3	-0.229	0.517	3.18	7	1.000	0.499	11.61	0.0861	0.740	2
world/air	2808	32763	0.0042	23.3	135.0	-0.081	0.518	3.61	10	0.999	0.651	66.30	0.0151	0.803	53
world/air+land	3223	39381	0.0038	24.4	124.7	-0.007	0.539	3.62	10	0.999	0.654	67.21	0.0149	0.825	54

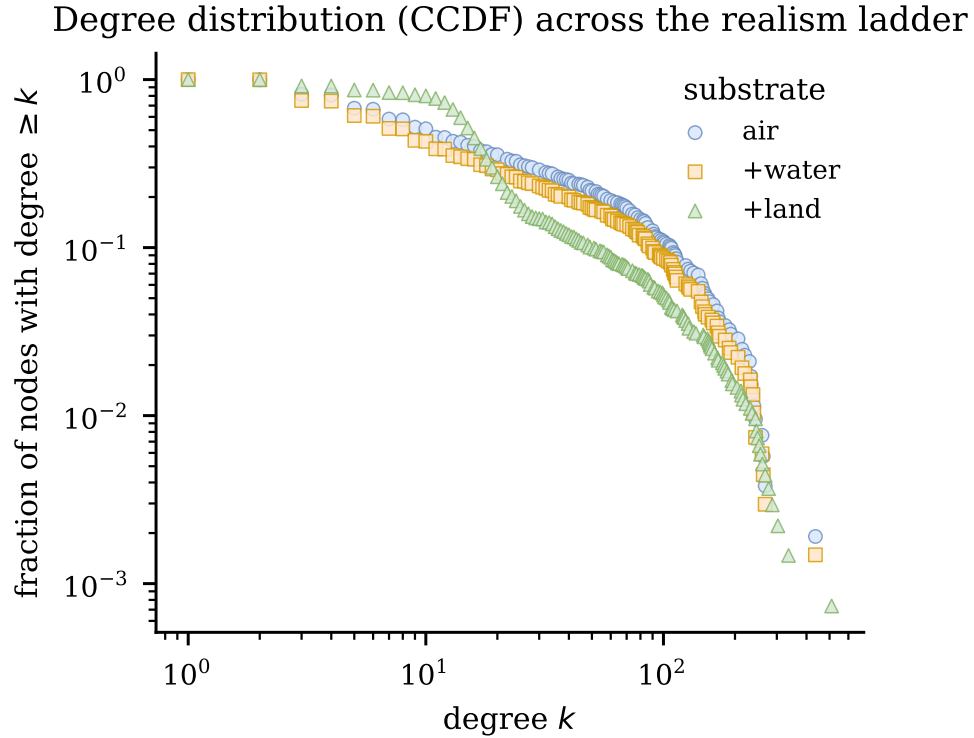


Figure 12: Degree distribution up the realism ladder, plotted as the complementary CDF (CCDF): each point is the fraction of cities with degree at least k , on log-log axes (the standard heavy-tail view, free of the $1/N$ tail noise of a raw histogram). Heavy-tailed over two orders of magnitude on every substrate; the land layer adds a shoulder of moderate-degree inland cities near $k \approx 12$ (Section 3.1).

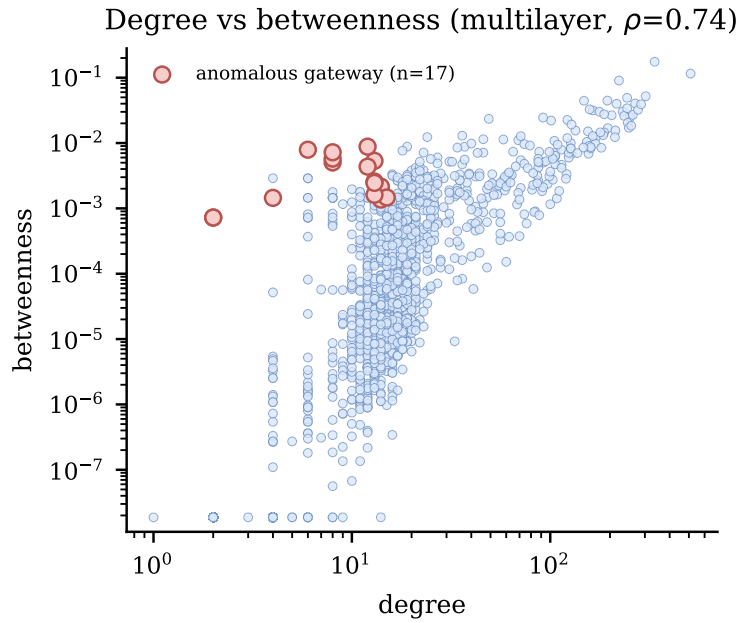


Figure 13: Degree versus betweenness per city on the multilayer network (log-log, $\rho = 0.74$). The two rank together for most cities, but the ringed anomalous gateways carry high betweenness at low degree: what betweenness-targeting catches and degree-targeting misses (Section 3.1).

C. IMPLEMENTATION ARCHITECTURE

The interactive simulator and the programmatic staged-protocol runs share one engine, configuration system, and results pipeline (Figure 14), so the demonstration and the study cannot drift apart.

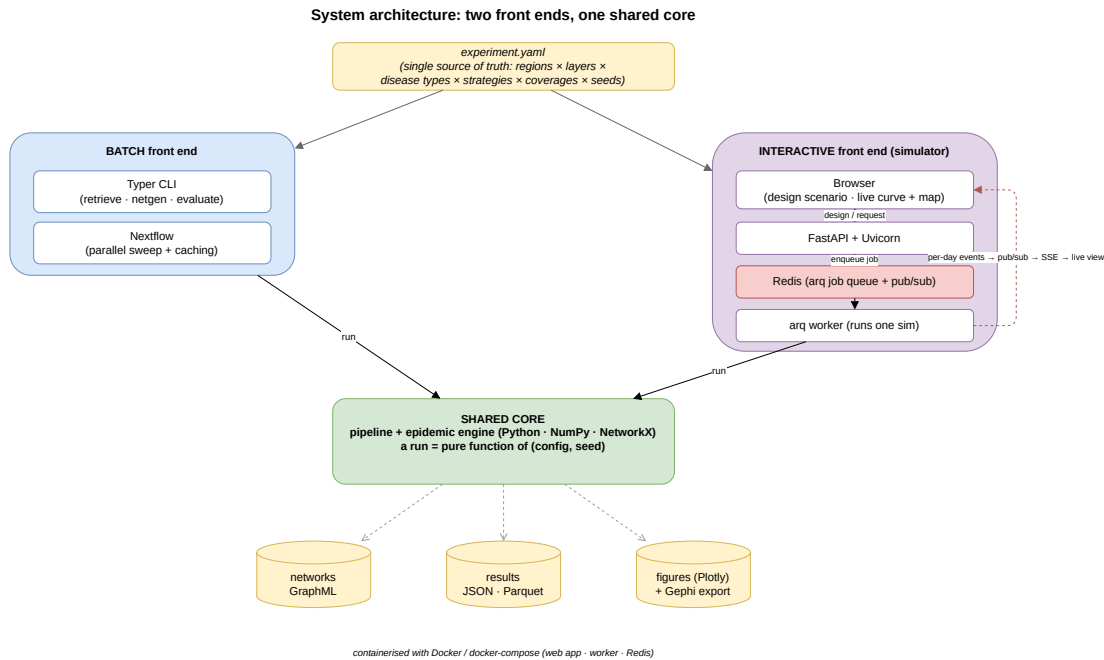


Figure 14: System architecture: one shared engine behind the interactive simulator and the programmatic runs, writing to a common results pipeline.